

PRAKTIKUM BASIS DATA

MODUL 5

Scripting Aggregate Function



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

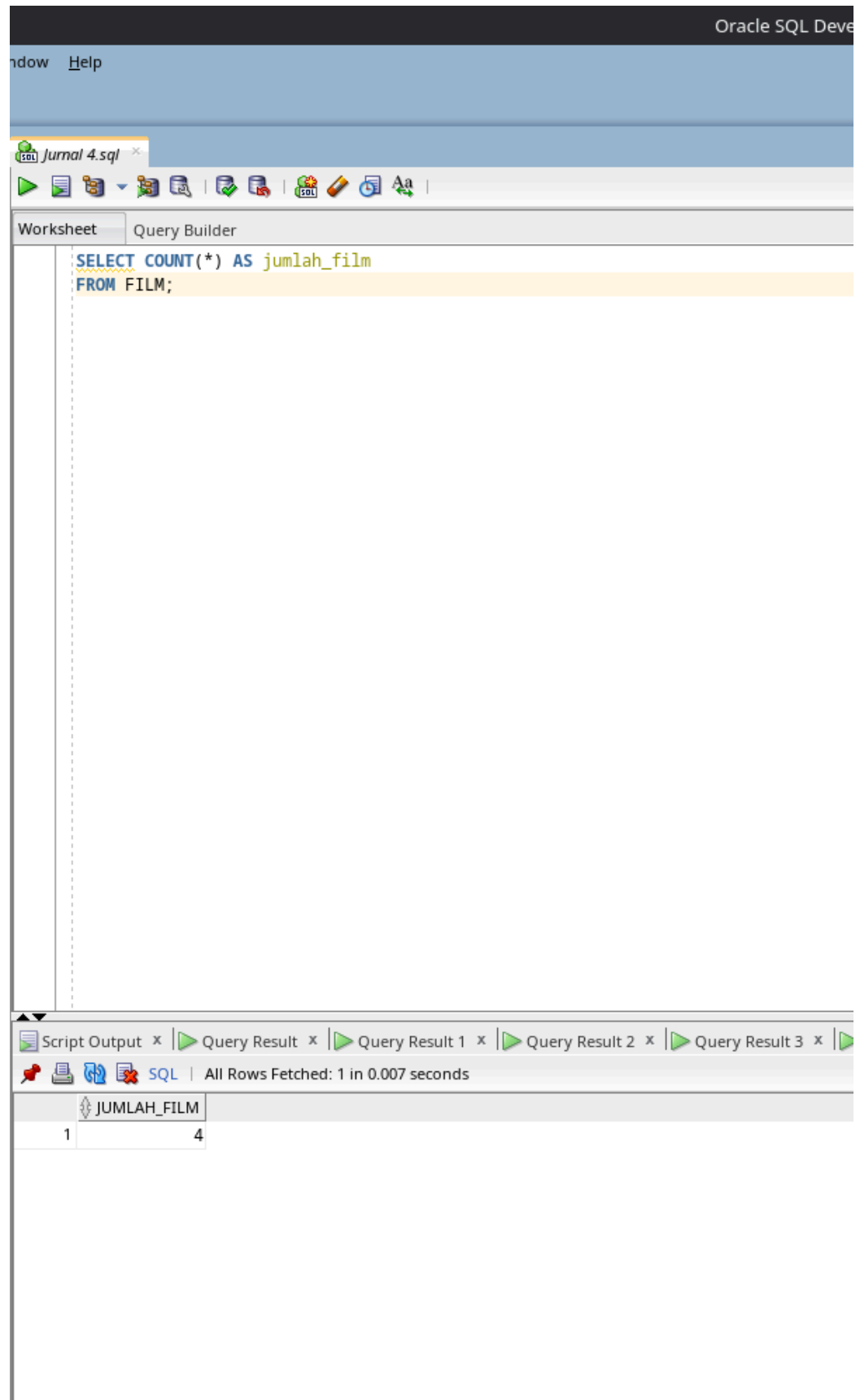
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TP

1. Kerjakan contoh kasus pada modul 5 (halaman 61). Sertakan Screenshot untuk setiap nomor.
 - a. Cari terlebih dahulu, ada berapa film yang tersedia di bioskop ini



- b. Kemudian, cari ada berapa film yang sedang diputar, dan satu judul hanya boleh dihitung satu kali

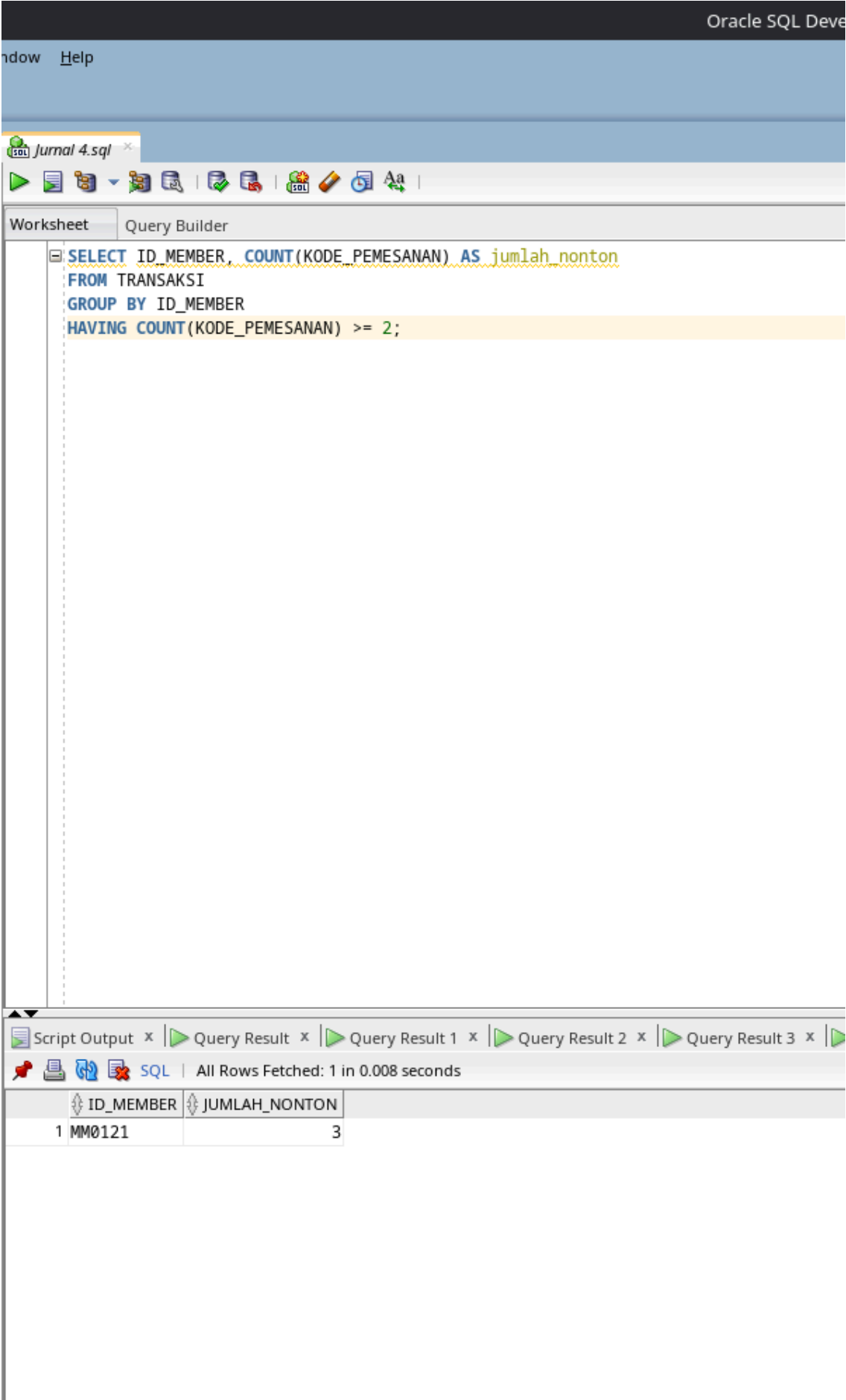
The screenshot shows the Oracle SQL Developer interface. The main window displays the following SQL query:

```
SELECT COUNT(DISTINCT f.ID_FILM) AS jumlah_film_tayang  
FROM FILM f  
JOIN JADWAL j ON f.ID_FILM = j.ID_FILM;
```

Below the query editor, the 'Script Output' tab is active, showing the query results. The results are displayed in a table with one column, 'JUMLAH_FILM_TAYANG', and one row with the value 4.

JUMLAH_FILM_TAYANG
4

- c. Cari id_member yang sudah menonton lebih dari 2 kali. Ulangi sekitar 3 sampai 5 kali



The screenshot shows the Oracle SQL Developer interface. The main window displays a SQL query in the Query Builder tab. The query is as follows:

```
SELECT ID_MEMBER, COUNT(KODE_PEMESANAN) AS jumlah_nonton
FROM TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(KODE_PEMESANAN) >= 2;
```

Below the query editor, the results pane shows the output of the query. It displays a table with two columns: ID_MEMBER and JUMLAH_NONTON. The results are as follows:

ID_MEMBER	JUMLAH_NONTON
1 MM0121	3

d. Cari member termuda dan jumlah film yang sudah ditontonnya.

The screenshot shows the Oracle SQL Developer interface. The main window displays an SQL query in the 'Worksheet' tab. The query is designed to find the youngest member and the number of movies they have watched. The query uses a subquery to identify the youngest member by selecting the maximum birth date from the 'MEMBER' table. The main query then joins this result with the 'TRANSAKSI' table to count the number of transactions (movies watched) for that member.

```
SELECT
  m.ID_MEMBER,
  m.NAMA,
  m.TGL_LAHIR,
  COUNT(tr.KODE_PEMESANAN) AS jumlah_film_ditonton
FROM MEMBER m
JOIN TRANSAKSI tr ON m.ID_MEMBER = tr.ID_MEMBER
WHERE m.TGL_LAHIR = (SELECT MAX(TGL_LAHIR) FROM MEMBER)
GROUP BY m.ID_MEMBER, m.NAMA, m.TGL_LAHIR;
```

Below the query editor, the 'Query Result' tab shows the output of the query. It indicates that all rows were fetched in 0.056 seconds. The result is displayed in a table with four columns: ID_MEMBER, NAMA, TGL_LAHIR, and JUMLAH_FILM_DITONTON. The table contains one row of data for member MM0122, named Dina, born on 20-AUG-01, who has watched 1 movie.

ID_MEMBER	NAMA	TGL_LAHIR	JUMLAH_FILM_DITONTON
1 MM0122	Dina	20-AUG-01	1

- e. Cari jumlah member yang lahir untuk setiap bulannya.

The screenshot shows the Oracle SQL Developer interface. The main window displays a SQL script named 'Jurnal 4.sql' in the 'Query Builder' tab. The script is as follows:

```
SELECT EXTRACT(MONTH FROM TGL_LAHIR) AS bulan,  
       COUNT(ID_MEMBER) AS jumlah_member  
FROM MEMBER  
GROUP BY EXTRACT(MONTH FROM TGL_LAHIR)  
ORDER BY bulan;
```

Below the script, the 'Script Output' tab shows the results of the query. The status bar indicates 'All Rows Fetched: 6 in 0.017 seconds'. The results are displayed in a table with two columns: 'BULAN' and 'JUMLAH_MEMBER'.

	BULAN	JUMLAH_MEMBER
1	1	1
2	5	1
3	7	1
4	8	1
5	9	1
6	11	1

f. Berapa jumlah jadwal tayang per film pada periode aktif?

The screenshot shows the Oracle SQL Developer interface. The main window displays a SQL query in the 'Query Builder' tab. The query is as follows:

```
SELECT f.ID_FILM, f.JUDUL, COUNT(j.ID_JADWAL) AS jumlah_jadwal
FROM FILM f
LEFT JOIN JADWAL j ON f.ID_FILM = j.ID_FILM
GROUP BY f.ID_FILM, f.JUDUL
ORDER BY jumlah_jadwal DESC;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the results of the query, which are displayed in a table with 4 rows and 3 columns: ID_FILM, JUDUL, and JUMLAH_JADWAL.

	ID_FILM	JUDUL	JUMLAH_JADWAL
1	F0102	Habibie Ainun 3	5
2	F0104	Buya Hamka	1
3	F0103	Taufiq	1
4	F0101	Keluarga Cemara	1

2. Buatlah table perbedaan antara Agregate Function dan Scalar Function. Berikan penjelasan dan contoh nya.

Fitur	Aggregate Function (Fungsi Agregat)	Scalar Function (Fungsi Skalar)
Definisi Utama	Fungsi yang mengolah sekumpulan data (banyak baris) dan menghasilkan satu nilai ringkasan	Fungsi yang mengolah satu nilai per baris dan menghasilkan satu nilai juga
Input	Banyak baris (multiple rows)	Satu nilai per baris
Output	Satu hasil untuk satu grup data	Satu hasil untuk setiap baris
Penggunaan Umum	Digunakan untuk menghitung total, rata-rata, jumlah, dll	Digunakan untuk manipulasi data seperti string, tanggal, atau angka
Lokasi Penggunaan	Biasanya dipakai dengan GROUP BY atau HAVING	Bisa dipakai di SELECT, WHERE, ORDER BY
Contoh Fungsi	SUM(), AVG(), COUNT(), MAX(), MIN()	UPPER(), LOWER(), ROUND(), LENGTH(), SYSDATE
Analogi	Seperti menghitung total nilai satu kelas	Seperti mengubah nilai tiap siswa satu per satu